



Iván Cárdenas

Geospatial Specialist · PhD Candidate

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SUMMARY

I am a Geospatial (GIS) Specialist with 8 years of experience in spatial data pipelines, Urban Digital Twin platforms, and production-grade Web-GIS applications for urban planning and environmental impact assessment. My experience spans Territorial Development, Cartography for Land Use Planning, Geodatabase Structuring, Hydrological Modeling, Spatial Analysis, Remote Sensing, Environmental Impact Assessment, and Research. Currently I am a PhD Candidate in the Urban Planning and Management Department at the University of Twente with a research project focused on Digital Twins as an enhanced 3D Planning Support System.

EXPERIENCE

PhD Candidate

ITC — University of Twente, Enschede, NL · 10/2023 - Present · 2 yrs 6 mo

Urban Digital Twins · DPSIR framework · Directed Acyclic Graphs · Python · GeoDjango · PostGIS · React · TiTiler · ArcGIS · Claude Code

- ▶ **Built** a cross-sectoral (water supply, heat stress, housing) **Urban Digital Twin** — open-source platform (GeoDjango + PostGIS + React + TiTiler) replacing manual SQL workflows via a DAG-driven spatial query builder. *To be published*
- ▶ **Delivered** Vitens' Water Supply Decision Support System (Python/Panel WebApp) modelling supply security, environmental impacts, and costs across multiple future scenarios.
- ▶ **Published** four peer-reviewed papers in top-tier journals advancing Decision Support Systems for Urban Planning.
- ▶ **Released** an open-source bibliometric enrichment tool for BibTeX metadata enabling seamless data integration across multiple academic databases.
- ▶ **Supervised** MSc students on research methods and technical skills.

Student Assistant

ITC — University of Twente × Vrije Universiteit Amsterdam, Enschede, NL · 01/2023 - 08/2023 · 8 mo

Urban Heat Modeling · ArcGIS · Python · Arcpy · Experience Builder

- ▶ **Designed and developed** a digital tool measuring Urban Heat Island effects and Perceived Body Temperature within a 3D environment under dynamic scenarios.
- ▶ **Built** a sequential geoprocess pipeline for iterative Urban Heat analysis under scenario simulation.
- ▶ **Designed** the user interface and cartographic styling for intuitive user interaction.

Natural Capital Intern

Koninklijke Philips N.V., Amsterdam, NL · 04/2023 - 08/2023 · 5 mo

Web-GIS · Mapbox GL · HTML · CSS · JavaScript

- ▶ **Developed** an interactive Web-GIS map of global manufacturing sites, integrating land-use and biodiversity data from remote sensing and external APIs using Mapbox GL, HTML, CSS, and JavaScript.
 - ▶ **Synthesised** multi-source satellite imagery and corporate environmental datasets into a unified spatial layer, replacing a manual spreadsheet workflow.
 - ▶ **Created** novel mapping solutions and data visualisations to communicate sustainability impact across all manufacturing sites.
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Geographic Information System Analyst

Freelancer, Bogotá, CO · 03/2016 - 08/2021 · 5 yrs 5 mo

Clients: Terrasos · ISA Intercolombia · Anglogold Ashanti · Ingema Limitada · MUZO Emerald Colombia · Acerías Paz del Río · Gran Tierra Energy · Transportadora de Gas Internacional · Cundinamarca Regional Government · Duitama City Hall

Geospatial Consultant for 10 major clients across urban planning, environmental impact assessment, and hydrological modelling (HEC-HMS, HEC-RAS) including Territorial Development plans and UAV imagery pipelines:

- Spatial analysis, geoprocess automation, and cartographic data visualisation.
- Geospatial database design, data collection, and quality assurance.
- Hydrological modelling, risk assessment, and environmental impact analysis.
- Remote sensing imagery processing and UAV (drone) mapping.
- Territorial planning and stakeholder engagement, translating spatial analysis into decision-ready reports.

EDUCATION

MSc Geo-Information Science and Earth Observation - Specialisation in Urban Planning

University of Twente · 09/2021 - 07/2023

Cum Laude · 9.0/10.0 GPA · ITC Excellence Scholarship 2021

Thesis: From Trash To Digital Treasure: Urban Digital Twinning for Solid Waste Management. A Case Study in the City of Tshwane, South Africa.

Recipient of the second-best Master's Thesis in Geoinformation in the Netherlands, 2023.

Postgraduate Specialisation — Geographical Information Systems

Universidad Distrital Francisco José de Caldas · 04/2019 - 11/2019

9.3/10.0 GPA · Top 5% of same program and academic period

Dissertation: Pilot Application for Real Estate Price Consultation and Quick Appraisal Based on Spatial Variables in the City of Bogotá.

Relevant coursework: Geospatial Database Design, Spatial Statistics, Remote Sensing Processing, Spatial Modeling and Analysis, User Interface Design.

BEng Environmental and Sanitary Engineering

Universidad de La Salle · 01/2009 - 04/2015

7.6/10.0 GPA · Top 7% of same program and academic period

Dissertation: Evaluation of Pulverised Rubber as a Raw Material in the Production of Roof Tiles as an Environmental Alternative for the Management of Used Tyres

TEACHING

Course Instructor

UNIGIS — Vrije Universiteit Amsterdam · 2024 - 2025

- Digital Twin, MSc Geographical Information Sciences — 2024, 2025

Guest Lecturer

University of Twente · 2024 - 2025

- Digital Twin Dashboard Workshop · 3D Modelling for City Digital Twins (2025)
- Urban Heat Modelling · Climate Transition in the IJssel Delta (2024-2025)

Teaching Assistant

University of Twente · 2024

- 3D Modelling for City Digital Twins
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PUBLICATIONS

ORCID: 0009-0005-0245-633X · University of Twente Research Profile

2026

BibTeX Enricher (*Software*) Cardenas-Leon, I. · 12 Mar 2026 Research output: Non-textual form › Software › Academic  Open Access

Cross-sectoral digital planning: Drinking Water–Urban Heat–Housing Provision indicators dataset Cárdenas-León, I., Nourian, P., Koeva, M. N. & Pfeffer, K. · Feb 2026 *Data in Brief*, 64, 17 p., 112310. Research output: Contribution to journal › Article › peer-review  Open Access

Digital Twin Dashboard Workshop (*Other contribution*) Cardenas-Leon, I. · 4 Jan 2026 Research output: Other contribution › Academic  Open Access

From planning question to spatial database: A geospatial data importer for urban digital twins (*Software*) Cardenas-Leon, I. · 12 Mar 2026 Research output: Non-textual form › Software › Academic  Open Access

2024

Digital twin-based framework for heat stress calculation Cárdenas-León, I., Morales-Ortega, R., Koeva, M., Atún, F. & Pfeffer, K. · 18 Oct 2024 *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, X-4-2024, 4, pp. 67–74. Research output: Contribution to journal › Conference article › peer-review  Open Access · 3 citations (Scopus)

Planning Support System Development for analysing environmental performance of form-based design decisions: Case Study of Enschede Tripathy, A., Koeva, M. N., Cárdenas-León, I. & Nourian, P. · 27 Jun 2024 In: *International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences — ISPRS Archives*, 48 (4/W11-2024), pp. 137–144. Research output: Chapter in Book/Report/Conference proceeding › Conference contribution › peer-review  Open Access

Solid Waste in the Virtual World: A Digital Twinning Approach for Waste Collection Planning Cárdenas, I., Koeva, M., Davey, C. & Nourian, P. · 21 Feb 2024 In: *Recent Advances in 3D Geoinformation Science — Proceedings of the 18th 3D GeoInfo Conference*. Kolbe, T. H., Donaubauer, A. & Beil, C. (eds.). Springer, pp. 61–74. (Lecture Notes in Geoinformation and Cartography). Research output: Chapter in Book/Report/Conference proceeding › Conference contribution › peer-review  Open Access · 8 citations (Scopus)

Urban digital twin-based solution using geospatial information for solid waste management Cárdenas-León, I., Koeva, M., Nourian, P. & Davey, C. · 15 Nov 2024 *Sustainable Cities and Society*, 115, 105798. Research output: Contribution to journal › Article › peer-review  Open Access · 22 citations (Scopus)

2023

Developments, challenges and applications of city digital twins in the context of the SDGs. Digital twins for a sustainable world Koeva, M. N., Cardenas, I. L. & Morales, L. R. · 1 Oct 2023 *GIM International*, 37(7), pp. 26–29. Research output: Contribution to journal › Article › Professional  Open Access

Digital Twins for Physiological Equivalent Temperature Calculation Guide (*Other contribution*) Cárdenas, I. L., Morales, R., Koeva, M., Atun, F. & Pfeffer, K. · 31 Aug 2023 Zenodo. Research output: Other contribution › Academic  Open Access

2022

Mapping ecosystem service flows in Senegal: Sardinella protein flows from small-scale fisheries (*Presentation*) Cárdenas-León, I., Sabu, L. & Awuor, E. · 21 Apr 2022 Zenodo. Research output: Other contribution › Academic  Open Access

AWARDS

YEAR	AWARD	ORGANISATION
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2025	Winner - Code Verification Competition	University of Twente
2023	Second Place — Best Master Thesis on Geo-Information Sciences	Geo-Informatie Nederland (GIN)
2021	Winner — Digital Twin Hackathon	University of the Witwatersrand, Johannesburg
2020	Winner — Map Gallery Best Cartography	ESRI Colombia

SKILLS

Expert · Advanced · Intermediate · Basic					
Skill		Level	Skill		Level
Geospatial					
ArcGIS			QGIS		
ESRI Suite			GeoServer		
PostGIS			GeoDjango		
TITiler			Mapbox GL JS		
HEC-RAS			HEC-HMS		
CesiumJS					
Backend & Data					
Python			Django		
PostgreSQL			Pandas		
GeoPandas			Streamlit		
Holoviz Panel					
Frontend					
JavaScript			HTML5		
CSS3			Tailwind CSS		
Design & Viz					
Cartographic Styling			Adobe Illustrator		
InDesign			Blender 3D		
Others					
Git			GitHub Pages		
Shopify					

LANGUAGES

LANGUAGE	LEVEL
Spanish	Native
English	C2 - Fluent
Portuguese	B1 - Intermediate
Dutch	A2 - Basic